Sectors

SEE ALSO
❮ 56–59 Ratio and proportion
❮ 138–139 Circles
❮ 140–141 Circumference and diameter

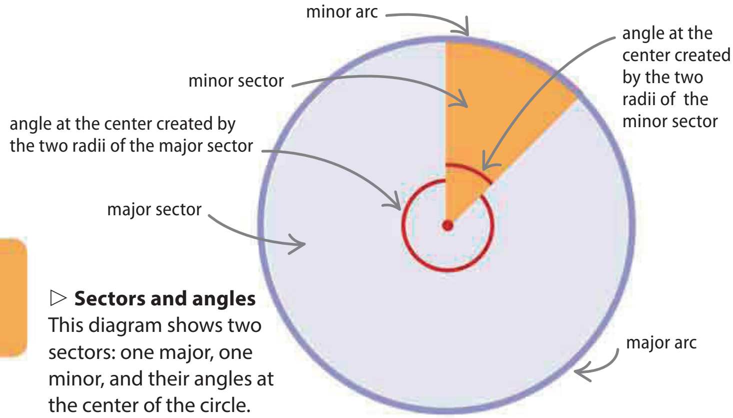
A SECTOR IS A SLICE OF A CIRCLE'S AREA. ITS AREA CAN BE FOUND USING THE ANGLE IT CREATES AT THE CENTER OF THE CIRCLE.

What is a sector?

A sector of a circle is the space between two radii and one arc. The area of a sector depends on the size of the angle between the two radii at the center of the circle. If the area of a sector is unknown, it can be found using this angle and the area of the circle. When a circle is split into two sectors, the bigger is called the "major" sector, and the smaller the "minor" sector.

$$\frac{\text{area of sector}}{\text{area of circle}} = \frac{\text{angle at center}}{360^\circ}$$

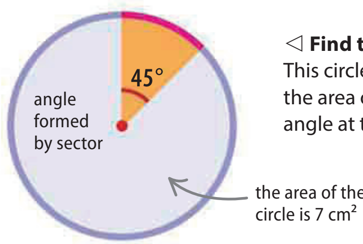
formula for finding the area of a sector



▷ **Sectors and angles** This diagram shows two sectors: one major, one minor, and their angles at the center of the circle.

Finding the area of a sector

The area of a sector is a proportion, or part, of the area of the whole circle. The exact proportion is the ratio of the angle formed between the two radii that are the edges of the sector and 360º. This ratio is part of the formula for the area of a sector.



◁ **Find the sector area** This circle has an area of 7 cm². Find the area of the sector that forms a 45° angle at the center of the circle.

Take the formula for finding the area of a sector. The formula uses the ratios between the area of a sector and the area of the circle, and between the angle at the center of the circle and 360º.

$$\frac{\text{area of sector}}{\text{area of circle}} = \frac{\text{angle at center}}{360^\circ}$$

total number of degrees in circle

Substitute the numbers that are known into the formula. In this example, the area is known to be 7 cm², and the angle at the center of the circle is 45º. The total number of degrees in a circle is 360º.

$$\frac{\text{area of sector}}{7} = \frac{45}{360}$$

this side has been multiplied by 7 to leave the area of a sector on its own (÷7 ×7 cancels out)

this side has also been multiplied by 7

Rearrange the equation to isolate the unknown value—the area of the sector—on one side of the equals sign. In this example, this is done by multiplying both sides by 7.

$$\text{area of sector} = \frac{45 \times 7}{360}$$

Multiply 45 by 7 and divide the answer by 360 to get the area of the sector. Round the answer to a suitable number of decimal places.

0.875 is rounded to 2 decimal places

$$C = 0.88\ \text{cm}^2$$